19 Two charges $+Q$ and $-2Q$ are placed at a distance $2d$ apart. Point X is at a distance d away from the $-2Q$ charge as shown below:



What is the electrical potential energy of a $+3Q$ charge placed at X?

A $\dfrac{5Q^2}{4\pi\varepsilon_0 d}$ **B** $\dfrac{-5Q^2}{4\pi\varepsilon_0 d}$ **C** $\dfrac{-Q^2}{8\pi\varepsilon_0 d^2}$ **D** $\dfrac{Q^2}{12\pi\varepsilon_0 d^2}$

Ans: B

Electrical potential at X, $V_X = \dfrac{Q}{4\pi\varepsilon_o(3d)} + \dfrac{-2Q}{4\pi\varepsilon_o(d)} = \dfrac{-5Q}{12\pi\varepsilon_o d}$

Electrical potential energy required, $U = (3Q)\left(\dfrac{-5Q}{12\pi\varepsilon_o d}\right) = \dfrac{-5Q^2}{4\pi\varepsilon_o d}$